

From Predictability to Tradability: A Comparative Study of Image- and Sequence-based Deep Learning in Cryptocurrency Markets

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Abstract

We compare image-based models (ResNet, DenseNet, Vision Transformer) and time-series models (LSTM, GRU, Transformer) for cross-sectional prediction of cryptocurrency returns under a unified experimental design. Using OHLCV data from five rolling study periods from 2018 to 2024, 30-day lookbacks, and forecast horizons of 1, 7, 14, and 28 days, model scores are mapped into equal-weight long-short portfolios via horizon-matched tranche construction, with transaction costs of 15 basis points per half-turn. Deep learning models learn weak but genuine cross-sectional signals. On traded assets, several model-horizon combinations exceed a 0.50 directional hit-rate benchmark, and mean Spearman rank information coefficients increase with the forecast horizon. However, these statistical patterns do not reliably translate into robust economic value once realistic trading frictions are imposed. Net Sharpe ratios are generally negative or small, and pairwise comparisons reveal no statistically significant winner after false discovery rate control. Image-based models exhibit lower volatility, but this does not translate into superior risk-adjusted performance. Robustness checks across study periods and portfolio breadths confirm that turnover and transaction costs are binding constraints on economic value. Overall, the results document a clear predictability-tradability gap in cryptocurrency markets.

Keywords: Cryptocurrency markets, Deep learning, Cross-sectional return prediction, Portfolio-based evaluation, Trading frictions

1. Introduction

Over the past decade, the cryptocurrency market has expanded rapidly in both market capitalization and the variety of instruments. While originally conceived as a decentralized payment system, it has evolved into a heterogeneous asset class characterized by high volatility, episodic regime shifts, and substantial cross-sectional dispersion. Persistent and large price fluctuations in major cryptocurrencies such as Bitcoin and Ethereum have continued to motivate research on predictability and market efficiency [17]. Much of the early literature on weak-form efficiency in cryptocurrency markets relied on linear time-series models, such as AR and ARIMA specifications, which are ill-suited

to environments marked by nonlinearity, structural breaks, and nonstationarity. In contrast, machine learning and deep learning offer flexible tools capable of capturing complex nonlinear patterns without restrictive functional assumptions, and have shown promise in short-horizon return prediction [20, 1, 13, 4]. Related evidence from equity markets further suggests that modern machine learning methods can extract economically relevant predictive structure from high-dimensional financial data [8], motivating their application to digital asset markets.

Despite this growing literature, relatively little attention has been paid to how data representation itself shapes the learning behavior of deep models in financial prediction tasks. In most applications, the same underlying open, high, low, close prices, and volume (OHLCV) information is fed into different architectures with limited discussion of how alternative representations—numerical sequences versus visual encodings—affect what models learn and how learned signals behave across horizons. This paper takes representation choice as its starting point. We ask whether transforming identical OHLCV data into sequence-based inputs or image-based chart representations leads deep learning models to extract qualitatively different predictive structures, and how these differences manifest in both statistical measures of predictability and portfolio-level outcomes.

By holding the underlying OHLCV information, forecast horizons, portfolio mapping, and trading cost assumptions fixed, our design isolates the role of representation-induced inductive biases. This allows us to attribute differences in learned signals—such as cross-sectional ranking strength, signal stability, and portfolio volatility—to representation choice rather than to confounding differences in information sets or implementation.

To address this question, we systematically compare two deep learning paradigms for cross-sectional cryptocurrency forecasting within a unified experimental framework. The image-based approach converts rolling 30-day OHLCV windows into candlestick charts with aligned volume bars and inputs them to vision backbones such as ResNet, DenseNet, and the Vision Transformer (ViT) [9, 11, 5]. This approach builds on recent evidence that vision architectures can learn predictive visual patterns from price charts in equity markets [14]. The sequence-based approach, by contrast, processes structured numerical time series and employs LSTM, GRU, and Transformer architectures to model temporal dependencies directly [6, 23]. By holding the underlying information set fixed and varying only the representation and the associated inductive biases of the models, we can isolate how representation choice shapes learning behavior across forecasting horizons.

Our objectives are twofold. First, we assess how representation (image versus sequence) and model architecture interact with the forecast horizon $h \in \{1, 7, 14, 28\}$ to shape statistical predictability, measured through portfolio-level directional signals and cross-sectional ranking performance. A key design choice in our study is to formulate the prediction task as a directional classification problem rather than a return regression. This choice is deliberate and reflects both economic and data-science considerations. In cryptocurrency mar-

kets, returns are heavy-tailed, highly volatile, and subject to frequent regime shifts, rendering point forecasts of expected returns unstable and overly sensitive to outliers. Our objective is therefore not to estimate precise return magnitudes, but to learn reliable cross-sectional orderings that support relative long-short decisions. A classification-based formulation, combined with a neutral band around zero, mitigates label noise from small price fluctuations and yields more stable ranking signals that are directly aligned with portfolio construction. Second, we evaluate whether the predictive structure learned under different representations translates into economically meaningful long-short portfolios once realistic trading frictions are imposed. To ensure comparability between predictive signals and their implementation, we map model outputs to horizon-matched tranche portfolios with equal-dollar weights, explicitly deduct transaction costs of 15 basis points per half-turn, and evaluate performance over five rolling study periods with a 30-day lookback. These design choices align statistical evaluation with tradable outcomes and reflect the frictions documented in cryptocurrency markets [18]. Accordingly, our objective is not to maximize predictive accuracy per se, but to learn stable cross-sectional orderings that are robust to noise and directly aligned with implementable portfolio decisions.

A priori, high turnover and transaction costs are known to erode apparent profitability in high-frequency and high-churn strategies[15, 21], and extensive model comparison raises concerns about data snooping and multiple testing[2, 10]. Our experimental design explicitly incorporates these considerations by adopting horizon-matched portfolio construction, conservative cost assumptions, and adjusted statistical inference.¹

Using this framework, we find that deep learning models extract weak but genuine cross-sectional signals from cryptocurrency markets. On traded assets, several model-horizon combinations exceed a 0.50 directional hit-rate benchmark, and mean Spearman rank information coefficients are positive, particularly at longer horizons where sequence-based Transformer models are strongest. However, these statistical patterns do not reliably translate into robust economic value once trading frictions are taken into account. Net risk-adjusted performance is generally small or negative, and no representation or architecture exhibits statistically significant dominance after multiple-testing correction. Image-based models tend to produce smoother portfolio trajectories and lower realized volatility, but without commensurate gains in risk-adjusted returns. Robustness checks across study periods and portfolio breadths confirm that turnover and transaction costs are binding constraints on economic value. These findings are consistent with the broader empirical asset pricing literature on frictions and post-discovery decay [19, 21] and with the fragmented, high-cost microstructure of cryptocurrency markets [18].

This study contributes to the literature in three ways. First, from a data science perspective, we introduce a controlled experimental design that isolates representation-induced inductive biases by comparing image-based and

¹We report adjusted p -values and use horizon-matched block bootstraps; see Section 2.

sequence-based deep learning models trained on identical OHLCV information, forecast horizons, portfolio mappings, and trading cost assumptions. Second, we propose a classification-to-ranking-to-portfolio pipeline that explicitly separates predictive learning from economic decision-making. By formulating prediction as directional classification and evaluating signals through cross-sectional rankings and horizon-matched portfolios, our framework aligns learning objectives with implementable long-short decisions under realistic market frictions. Third, we empirically quantify and diagnose a predictability-tradability gap in cryptocurrency markets. While deep learning models extract weak but systematic cross-sectional signals, turnover and transaction costs emerge as binding constraints that prevent these signals from translating into robust economic value, even under favorable statistical metrics.

This paper is related to the literature on machine learning-based prediction and trading in cryptocurrency markets. Prior studies document that deep learning models such as LSTM and GRU can outperform linear benchmarks in predictive accuracy, but that performance is highly sensitive to forecast horizons, trading costs, and implementation choices [20, 1, 4, 13]. Reinforcement learning approaches have also been explored, reporting improvements over rule-based strategies under specific execution assumptions [3]. A parallel literature investigates alternative representations of financial time series, transforming price histories into visual inputs for convolutional or attention-based models, with strong results in equity markets [14]. However, image-based approaches remain relatively underexplored in cryptocurrency markets, and direct comparisons between image-based and sequence-based representations using identical OHLCV information are rare. More broadly, research in empirical asset pricing shows that machine learning can uncover nonlinear cross-sectional structure [8], while emphasizing that transaction costs, turnover, and post-discovery decay often erode apparent profitability [15, 21, 19]. In cryptocurrency markets, common risk factors such as market, size, and momentum have been documented [17, 16], motivating momentum strategies as benchmarks. Our study builds on this literature by jointly examining data representation and forecast horizon within a portfolio-based evaluation framework that explicitly accounts for trading frictions and robust statistical inference.

The remainder of the paper proceeds as follows. Section 2 describes the data, modeling choices, and experimental design. Section 3 reports empirical results and Section 4 discusses implications of empirical results. Section 5 concludes.

2. Methodology

This study follows a structured multi-stage empirical framework, building on and extending the approaches of Fischer et al. [7], Fischer and Krauss [6], Jaquart et al. [13]. The framework proceeds as follows. First, we collect OHLCV data and construct investable cross-sectional universes. Second, we transform the raw price and volume series into model-ready representations and horizon-specific prediction targets using a 30-day rolling lookback. Third, we implement rolling study periods with clearly separated training, validation, and test sets

and estimate the prediction models within each period. Finally, we evaluate both statistical predictability and economic performance using strictly out-of-sample portfolio backtests that explicitly incorporate trading frictions.

2.1. Data

We use daily cryptocurrency data denominated in U.S. dollars from February 1, 2018, to December 15, 2024, including open, high, low, and close prices, trading volume (OHLCV), and market capitalization. Prices are obtained from CryptoCompare’s CCCAGG index, which aggregates data across major exchanges using volume-weighted averaging and outlier controls. Daily observations are timestamped at 00:00 UTC and correspond to the previous calendar day’s close. To ensure a strict out-of-sample evaluation, all predictors are lagged by one trading relative to portfolio formation, and any scaling or normalization is performed using statistics estimation exclusively from the trading window. In particular, we define the daily *simple* return for coin c as

$$r_{t+1}^c = \frac{P_{t+1}^c}{P_t^c} - 1.$$

All labels, momentum measures, and backtests use simple returns r_t^c . We do not use log returns in model training or evaluation.

To construct a consistent and investable cross-sectional learning universe, we define, for each study period, the investable set as the top 100 cryptocurrencies ranked by market capitalization on the first day of the training window. Fiat-pegged stablecoins such as USDT, BUSD, and USDC are excluded. Once defined, the universe is held fixed throughout the corresponding study period, even if assets subsequently delist or decline in capitalization. This design choice serves two data-science-motivated purposes. First, it mitigates survivorship and look-ahead biases by preventing dynamic universe reconstruction based on future information. Second, it provides a stable cross-sectional domain for representation learning, ensuring that differences in predictive performance across models are not mechanically driven by changes in asset composition. Assets with missing or non-finite inputs are excluded on the affected dates, and the set of tradable assets on day t is denoted by $\mathcal{C}(t)$.

Given the pronounced non-stationarity and regime dependence of cryptocurrency markets, we adopt a rolling study-period design. Specifically, the full sample is divided into five overlapping study periods (SP1–SP5), each spanning 1,350 trading days and consisting of a training window (810 days), a validation window (270 days), and a strictly held-out test window (270 days). Each subsequent study period is shifted forward by one test window, producing non-overlapping out-of-sample segments. This construction allows us to evaluate model performance across heterogeneous market regimes, ranging from bull markets to sharp drawdowns, while preserving a clean separation between model estimation, hyperparameter tuning, and evaluation. Each study period can be viewed as a local data-generating regime, and models are trained independently within each regime to reduce implicit pooling across structurally different environments. The training and validation sets jointly form the *formation window*

used to fit scalars and tune hyperparameters. The test sets are held out strictly for evaluation. Table 1 reports the exact calendar ranges.

Table 1: Study periods and date ranges for training, validation, and test sets.

SP No	Training Set	Validation Set	Test Set
1	2018-05-02 – 2020-07-09	2020-07-10 – 2021-04-05	2021-04-06 – 2021-12-31
2	2019-01-27 – 2021-04-05	2021-04-06 – 2021-12-31	2022-01-01 – 2022-09-27
3	2019-10-24 – 2021-12-31	2022-01-01 – 2022-09-27	2022-09-28 – 2023-06-24
4	2020-07-20 – 2022-09-27	2022-09-28 – 2023-06-24	2023-06-25 – 2024-03-20
5	2021-04-16 – 2023-06-24	2023-06-25 – 2024-03-20	2024-03-21 – 2024-12-15

Within each study period, we construct model inputs using a 30-day rolling lookback window. We consider two distinct but parallel representation pipelines. The first is a sequence-based numerical representation, in which daily percentage changes of OHLC prices and log-differences of trading volume are computed for each asset. Specifically, for each coin c and day t , we compute one-day percentage changes of OHLC and the log-difference of volume, such as,

$$\Delta p_t^c = \frac{p_t^c - p_{t-1}^c}{p_{t-1}^c}, \quad p \in \{\text{Open, High, Low, Close}\}, \quad \ell V_t^c = \log(1 + V_t^c),$$

$$\Delta \ell V_t^c = \ell V_t^c - \ell V_{t-1}^c = \log \frac{1 + V_t^c}{1 + V_{t-1}^c}.$$

These transformations render the inputs approximately scale-invariant across assets and time, facilitating cross-sectional learning by deep sequence models. The resulting multivariate time series are stacked into fixed-length windows and standardized using only the training-sample statistics. This representation preserves temporal ordering and magnitude information while avoiding raw price-level nonstationarity. Observations with missing or non-finite inputs are removed prior to scaling.²

The second pipeline is an image-based visual representation, in which each 30-day window is rendered as a candlestick chart with aligned volume bars. Importantly, all charts are generated using fixed axes within the window, without future-aware autoscaling or auxiliary annotations. Figure 1 shows an example of a 30-day candlestick chart. This constraint is imposed to prevent implicit information leakage and to ensure that the visual encoding reflects only information available up to the prediction time. This transformation maps structured numerical data into an unstructured visual manifold, enabling convolutional and attention-based vision architectures such as ResNet, DenseNet, and ViT to exploit spatial inductive biases and pattern abstraction that differ fundamentally from those of sequence models.

²Using log-differences for volume avoids division by nearly zero values and is invariant to rescaling; for small changes, it closely approximates percentage changes.

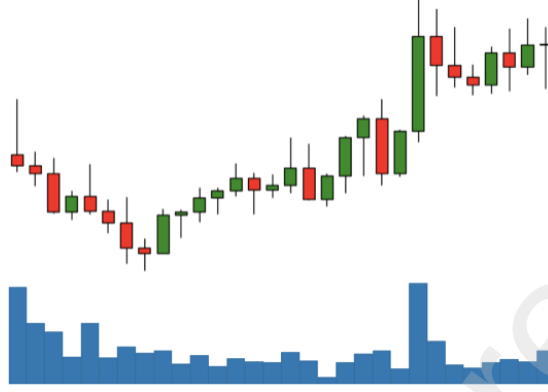


Figure 1: Example of a 30-day candlestick chart with volume bars used as image input. The chart is rendered with fixed axes and aligned volume bars for each window.

By construction, both representation pipelines are derived from the same underlying OHLCV data, share identical lookback lengths, and are aligned in time. This alignment ensures that any observed differences in predictive or economic performance can be attributed to the interaction between representation choice and model architecture, rather than to discrepancies in information content. Taken together, the data and representation design in this study establish a unified and controlled input space in which deep learning models with distinct inductive biases can be meaningfully compared under realistic financial market conditions.

2.2. Deep learning Prediction Models

Building on the unified data and representation framework described in Section 2.1, we formalize the prediction task and the associated deep learning models. We emphasize not only model architecture but also target definition, label engineering, and the mapping from raw model outputs to cross-sectional prediction signals. Our objective is not to identify a single best predictive model, but to examine how different architectural inductive biases interact with representation choice and forecast horizon in learning across cross-sectional structure.

We formulate cryptocurrency return prediction as a horizon-dependent directional classification problem. For each asset c and prediction date t , we define cumulative simple returns over horizons $h \in \{1, 7, 14, 28\}$ trading days ahead as

$$R_{t \rightarrow t+h}^c = \prod_{j=1}^h (1 + r_{t+j}^c) - 1,$$

where r_{t+j}^c denotes the one-day simple return. These horizons are chosen to span short-term, intermediate, and multi-week forecasting tasks, allowing us to examine how predictive content aggregates as noise is averaged out over longer

holding periods. From a learning standpoint, each horizon corresponds to a distinct prediction task with potentially different signal-to-noise characteristics and degrees of temporal persistence.

Rather than modeling raw returns directly, we discretize future outcomes into a three-class label space. Specifically, cumulative returns are mapped to upward, neutral, and downward classes using a symmetric tolerance band around zero. This neutral band is introduced to mitigate label noise arising from very small price movements, which are particularly prevalent in daily cryptocurrency data and can obscure the learning signal. Importantly, the neutral class is not interpreted as an economically inactive region, but as a label-engineering device that stabilizes learning and improves cross-sectional ranking by preventing marginal fluctuations from dominating the loss function.

Specifically, we use a $\pm 0.5\%$ band as our design choice which is consistent with prior three class stock direction setups and multimodal stock-movement tasks [e.g., 22, 25, 12, 24]. Let $\tau \equiv 0.005$. We then map returns to labels $y_{t,h}^c \in \{-1, 0, +1\}$ as

$$y_{t,h}^c = \begin{cases} +1 & \text{if } R_{t \rightarrow t+h}^c \geq \tau, \\ 0 & \text{if } |R_{t \rightarrow t+h}^c| < \tau, \\ -1 & \text{if } R_{t \rightarrow t+h}^c \leq -\tau. \end{cases}$$

Labels are defined on the forward window $[t+1, t+h]$, while all predictors use only information available up to day t , ensuring a clean temporal separation between inputs and targets. The features are lagged by one day and trades are supposed to execute at the close of t .

We consider two broad families of deep learning models corresponding to the representation pipelines introduced in Section 2.1, which are sequence-based models operating on numerical time series, and image-based models operating on visual chart representations. Within each family, we select architectures that are widely used in the machine learning literature and embody distinct inductive biases.

For sequence-based representations, we employ Long Short-Term Memory (LSTM) networks, Gated Recurrent Units (GRU), and Transformer models. LSTM and GRU architectures are designed to capture temporal dependencies through recurrent gating mechanisms, making them well-suited for modeling sequential patterns in the financial time series. The transformer, by contrast, replaces recurrence with self-attention, enabling flexible modeling of long-range dependencies and interactions across the entire lookback window. In the context of cross-sectional return prediction, these architectures differ in how they aggregate temporal information and how sensitive they are to regime changes and horizon length.

For image-based representations, we use ResNet, DenseNet, and Vision Transformer (ViT) architectures. Convolutional neural networks such as ResNet and DenseNet impose strong locality and hierarchical feature extraction through convolution filters, which can act as a form of regularization when learning from

noisy financial data. The Vision Transformer, in contrast, applies self-attention directly to image patches, allowing global visual patterns to be learned without predefined spatial locality. By including both convolutional and attention-based vision models, we are able to examine whether different visual inductive biases translate into systematically different predictive behavior in financial chart data.

All models are trained to output class probabilities for the three directional labels using cross-entropy loss. Hyperparameters are selected using the validation window within each study period, and early stopping is applied to prevent overfitting. Models are trained separately for each study period and each forecast horizon, reflecting the view that optimal representations and parameters may vary across market regimes and prediction tasks. This design avoids implicit pooling across structurally heterogeneous environments.³

While model training is framed as a classification problem, our primary object of interest is not classification accuracy per se, but the cross-sectional ordering of assets implied by model outputs. To this end, we transform predicted class probabilities into a scalar prediction score defined as the difference between the probabilities of upward and downward movements. This transformation yields an ordinal signal that is robust to calibration errors and is directly suited for ranking assets within the cross-section on each prediction date. This step represents a deliberate separation between predictive learning and decision-making. The scalar score is not interpreted as an estimate of expected return, but as a relative measure of directional confidence that facilitates cross-sectional comparison. By focusing on ranking rather than point prediction, we align the learning objective with the downstream portfolio construction problem and reduce sensitivity to distributional misspecification in the predicted probabilities.

2.3. Portfolio Construction and Economic Evaluation

We translate the cross-sectional prediction signals generated in Section 2.2 into investable portfolios and evaluate their economic performance under explicit trading frictions. This stage constitutes a decision layer that is deliberately separated from the prediction layer. Our aim is not to optimize portfolio performance ex post, but to assess whether statistically learned signals retain economic value once they are embedded in a transparent and constrained decision rule that reflects realistic trading conditions in cryptocurrency markets.

We benchmark against momentum strategies with lookbacks $L \in \{7, 14, 28\}$ days. The momentum score for the asset $c \in \mathcal{C}(t)$ at t is the past- L cumulative

³All features are lagged one day relative to trades, scalars are fit on the training window and applied out-of-sample, and model selection uses validation sets only. Image rendering uses fixed axes within the window and prohibits future-aware autoscaling or metadata that could leak information. We train separate models by SP and horizon to accommodate regime shifts. All hyperparameters and additional robustness such as alternative thresholds, k -breadths and cost grids are reported in the Appendix.

simple return (excluding day t):

$$M_t^c = \prod_{j=1}^L (1 + r_{t-j}^c) - 1.$$

Assets are sorted in descending order of scores, and portfolios are formed by taking equal-dollar long positions in the top k assets and equal-dollar short positions in the bottom k assets.

We obtain the score from the deep learning prediction. At the close of day t , each model produces, for every $c \in \mathcal{C}(t)$ and horizon h , a probability vector $\hat{\mathbf{p}}_{t,h}^c = (\hat{p}_{t,h}^{\text{up}}, \hat{p}_{t,h}^{\text{neu}}, \hat{p}_{t,h}^{\text{down}})$ using only information available up to $t-1$. Then, we use the scalar score $s_{t,h}^c = \hat{p}_{t,h}^{\text{up}} - \hat{p}_{t,h}^{\text{down}}$, and sort in descending order within $\mathcal{C}(t)$. Let $L_{t,h}$ and $S_{t,h}$ denote the top- k and bottom- k sets, respectively. Then we form equal-dollar long-short weights $w_{t,h}^c = +1/k$ for $c \in L_{t,h}$ and $w_{t,h}^c = -1/k$ for $c \in S_{t,h}$. Ties are resolved deterministically using market capitalization to ensure reproducibility. Throughout the analysis, signals and weights are fixed at t based on information up to $t-1$ and positions are executed at the close of t .

To ensure consistency between the prediction horizon and the investment horizon, we implement horizon-matched tranche portfolios. For a given forecast horizon h , a new tranche representing $1/h$ of total portfolio capital is initiated each day using the contemporaneous long-short ranking $(L_{t,h}, S_{t,h})$, while the tranche initiated h days earlier is simultaneously closed. As a result, the aggregate portfolio always consists of h overlapping tranches, each held to maturity without intermediate rebalancing.

The daily portfolio returns are computed as the average return across active tranches, net of transaction costs. Specifically, it is computed by

$$r_{p,t+1}^{(h)} = \frac{1}{h} \sum_{i=0}^{h-1} \left[\frac{1}{k} \sum_{c \in L_{t-i,h}} r_{t+1}^c - \frac{1}{k} \sum_{c \in S_{t-i,h}} r_{t+1}^c \right] - \text{TC}_{t+1}^{(h)},$$

Portfolio net asset value evolves multiplicatively based on daily simple returns, and no leverage scaling or volatility targeting is applied.

Transaction costs are incorporated explicitly and symmetrically for long and short positions. We assume proportional trading costs of 15 basis points per half-turn, applied to one-way turnover. In particular, let $w_{c,t}^{(h)} \in \{-1/k, 0, +1/k\}$ denote equal-dollar long-short weights for horizon h on day t , so the gross notional is $G_t = \sum_c |w_{c,t}^{(h)}| = 2$. Then, one-way turnover is defined by

$$\text{TO}_{t+1}^{(h)} \equiv \frac{1}{2} \sum_c |w_{c,t+1}^{(h)} - w_{c,t}^{(h)}|.$$

so that costs quoted “per half-turn” scale linearly with TO. Under horizon-matched tranches, a new $1/h$ tranche is opened and the tranche opened h days earlier is closed each day. Hence the baseline turnover equals $\text{TO}_{t+1}^{(h)} = \frac{2}{h}$, and

transaction costs are deducted from portfolio returns as

$$\text{TC}_{t+1}^{(h)} = \lambda \cdot \text{TO}_{t+1}^{(h)},$$

where $\lambda = 15$ bps per half-turn.

Remark 1. An equivalent bookkeeping counts openings and closings separately ($4/h$ one-way trades) with a per-half-turn fee of 7.5 bps. This produces identical TC and avoids double-counting under our convention. Positions within existing tranches are buy-and-hold until maturity, so no additional turnover arises within those tranches. In addition, we do not include stock-borrowing fees because shorting availability is inconsistent across the universe. Thus, the net asset value at $t + 1$ evolves as $\text{NAV}_{t+1} = \text{NAV}_t (1 + r_{p,t+1}^{(h)})$.

This explicit treatment reflects the fragmented and fee-intensive microstructure of cryptocurrency markets, where even liquid assets can face nontrivial execution costs. By fixing the cost parameter across all models, horizons, and study periods, we ensure that economic comparisons are not confounded by model-specific assumptions about execution quality. While the chosen cost level is conservative, it provides a stringent benchmark for assessing whether learned signals are economically meaningful under realistic implementation constraints.

We evaluate portfolio outcomes using both statistical learning metrics and economic performance metrics, with a clear conceptual distinction between the two. Statistical metrics include portfolio-level directional hit rates which is computed on traded assets only (top- k longs and bottom- k shorts) with three-class labels $(+1, 0, -1)$, and cross-sectional information coefficients (Spearman rank correlations) between prediction scores $\{s_{t,h}^c\}$ and realized returns (daily r_{t+1}^c or the h -day cumulative simple return $R_{t \rightarrow t+h}^c$, as appropriate). These measures assess the quality of learned rankings independently of trading frictions.

Economic performance is evaluated using cumulative returns, mean daily returns, annualized volatility, Sharpe and Sortino ratios, maximum drawdown, turnover, and tail-risk measures such as VaR (Value-at-Risk) and CVaR (Conditional VaR). All performance statistics are computed from daily simple returns, and annualization follows a standard 252-day convention, which provides

$$\text{Vol}_{\text{ann}} = \sqrt{252} \text{Vol}_{\text{daily}}, \quad \text{Sharpe}_{\text{ann}} = \sqrt{252} \text{Sharpe}_{\text{daily}}, \quad \text{Sortino}_{\text{ann}} = \sqrt{252} \text{Sortino}_{\text{daily}}.$$

For statistical significance, we test mean returns against zero and against momentum baselines with multiple-testing control.⁴

Taken together, the portfolio construction and evaluation framework is designed to expose the potential gap between predictability, as measured by statistical learning metrics, and tradability, as measured by net-of-transaction costs.

⁴Given the extensive model and horizon comparisons, statistical inference is conducted using paired block bootstrap procedures with block lengths matched to the forecast horizon. To control for data snooping and multiple testing, we apply family-wise and false discovery rate adjustments to all hypothesis tests. These procedures allow us to distinguish between apparent performance differences and statistically robust evidence of economic outperformance.

By holding the decision rules, cost assumptions, and evaluation procedures fixed across models, we provide a transparent assessment of how much of the predictive content extracted by deep learning models survives the transition from prediction to implementation.

3. Results

This section presents the empirical results following the structured methodology introduced in Section 2. The results are organized to evaluate deep learning models primarily through their implications at the portfolio level, and to assess whether the information embedded in model-generated signals translates into economically meaningful outcomes once realistic trading frictions are imposed.

To facilitate exposition, we begin by presenting detailed results from Study Period 3 (SP3) in Table 1, which serves as a representative out-of-sample environment. SP3 spans a period characterized by pronounced volatility, regime transitions, and gradual trend formation, making it particularly informative for evaluating how alternative data representations and model architectures perform across short-, medium-, and longer-horizon prediction tasks. Importantly, the qualitative patterns observed in SP3 closely mirror those found in the other study periods. Accordingly, SP3 provides a convenient and illustrative benchmark for discussing model behavior, while results from the remaining study periods are used to assess robustness and generality. All metrics are computed on daily returns as in Section 2. Unless otherwise stated, we use a representative portfolio breadth of $k = 5$ (top/bottom five). The additional k -breadths and other SPs are deferred to the Appendix.

3.1. Portfolio Performance Using RNN and CNN Models

We begin the analysis of results by examining portfolio-level performance in SP3, which provides the primary evidence on whether deep learning models extract economically relevant information from cryptocurrency markets. Consistent with the prediction decision framework outlined in Section 2, model outputs are evaluated not at the level of individual asset predictions, but through their implications for cross-sectional long-short portfolios constructed using model-based rankings.

Table 2 reports portfolio directional hit rates for SP3 across model architectures and forecast horizons. Hit rates are computed on traded assets only, corresponding to the long and short positions selected by each model-based ranking, with neutral outcomes treated as incorrect classifications. As such, this metric provides a coarse but intuitive measure of portfolio-level directional predictability, capturing whether model signals correctly anticipate the sign of subsequent returns on positions that actually enter the portfolio.

Several patterns emerge from Table 2. First, hit rates cluster around the 0.5 benchmark across models and horizons, indicating that none of the deep learning architectures deliver strong directional accuracy at the individual asset level. This result underscores the difficulty of short-horizon return prediction

Table 2: Portfolio directional hit rate (Top/Bottom- k) by model and horizon for SP3 ($k = 5$). Computed only on traded names; neutrals treated as incorrect.

Model	1D	7D	14D	28D
LSTM	0.4336	0.4436	0.4850	0.5200
GRU	0.4672	0.4821	0.4800	0.4900
Transformer	0.4269	0.5000	0.5050	0.4400
ResNet	0.4283	0.4487	0.5000	0.4500
DenseNet	0.4346	0.4487	0.4500	0.4600
ViT	0.4312	0.4487	0.5050	0.5000
Momentum(7)	0.4430	0.4703	0.4732	0.4646
Momentum(14)	0.4444	0.4728	0.4649	0.4562
Momentum(28)	0.4412	0.4672	0.4619	0.4458

in cryptocurrency markets, even when flexible nonlinear models are employed. At the same time, a number of model-horizon combinations modestly exceed the 0.5 threshold, particularly at longer horizons. For example, the LSTM model achieves a hit rate of 0.520 at the 28-day horizon, while the Transformer and Vision Transformer (ViT) reach values slightly above 0.50 at intermediate horizons. Momentum-based benchmarks, by contrast, remain consistently below 0.5 across all horizons. Taken in isolation, these magnitudes should not be interpreted as evidence of economically meaningful predictability; rather, they suggest the presence of weak directional structure that may only become informative when aggregated at the portfolio level.

To more directly assess the quality of cross-sectional signals, Table 3 complements the hit-rate analysis by reporting cross-sectional rank information coefficients (ICs), defined as the Spearman correlation between model-generated scores and subsequent realized returns.⁵ Unlike directional hit rates, ICs directly evaluate the ordinal quality of prediction signals across the investable universe and are therefore particularly informative for ranking-based portfolio construction.

The IC results provide clearer evidence of systematic cross-sectional learning. Across most deep learning models, ICs tend to increase with the forecast horizon, consistent with the notion that longer holding periods allow transient noise to average out and reveal more persistent ordering information. This pattern is particularly pronounced for the sequence-based Transformer, which attains ICs of 0.0845 at the 14-day horizon and 0.0942 at the 28-day horizon in SP3. Image-based models also exhibit positive ICs at longer horizons, although their magnitudes are generally smaller. In contrast, momentum benchmarks yield

⁵On each test date and for $h \in \{1, 7, 14, 28\}$, we compute the Spearman correlation between the cross-sectional ranks of $s_{t,h}^c = \hat{p}_{t,h}^c(\text{up}) - \hat{p}_{t,h}^c(\text{down})$ and the h -ahead simple return $R_{t \rightarrow t+h}^c$ (for $h=1$, $R_{t \rightarrow t+1}^c = r_{t+1}^c$), and report the time-series mean over the SP3 test window.

Table 3: Mean cross-sectional Spearman information coefficients (IC) between the score $s = \hat{p}_{\text{up}} - \hat{p}_{\text{down}}$ and h -ahead simple returns in SP3. ICs are computed daily in the cross-section and averaged over the test window.

Model	1D	7D	14D	28D
LSTM	0.0238	0.0059	-0.0017	0.0119
GRU	0.0675	0.0545	0.0424	0.0403
Transformer	0.0193	0.0493	0.0845	0.0942
ResNet	0.0281	0.0242	0.0167	0.0358
DenseNet	0.0263	0.0245	0.0402	0.0646
ViT	0.0269	0.0287	0.0597	0.0517
Momentum(7)	-0.0056	-0.0009	0.0119	0.0114
Momentum(14)	-0.0158	-0.0047	0.0017	0.0056
Momentum(28)	-0.0165	-0.0148	-0.0122	-0.0185

ICs that are close to zero or negative across horizons, indicating limited ability to rank assets in a cross-sectional setting using simple trend-following rules.

Taken together, Tables 2 and 3 highlight a key empirical insight of the paper. While deep learning models do not achieve strong asset-level directional accuracy, their portfolio-level signals nevertheless embed weak but systematic cross-sectional information. This distinction between limited pointwise predictability and nontrivial ranking quality is crucial for understanding the subsequent economic analysis. In the next subsection, we examine whether this portfolio-level predictability translates into economically meaningful performance once trading frictions and turnover are explicitly taken into account.

3.2. Economic Performance

We next turn to the economic performance of the portfolio strategies implied by deep learning signals in SP3. While Section 3.1 establishes that model-based rankings contain weak but systematic cross-sectional information, the more important question is whether this portfolio level predictability survives the transition from prediction to implementation once trading frictions are taken into account. To address this question, we evaluate horizon-matched, equal-dollar long-short portfolios by incorporating mechanically induced turnover and proportional transaction costs.

Table 4 summarizes the net economic performance of horizon-matched long-short portfolios constructed from deep learning model rankings in SP3. All performance statistics are computed from daily portfolio returns net of 15bps per half-turn transaction costs, using the equal-dollar weighting and tranche-based implementation. Panel A through D correspond to forecast horizons of 1, 7, 14, and 28 days, respectively, and report mean returns, annualized volatility, risk-adjusted performance measures, and tail-risk statistics. Momentum-based benchmark strategies are included for comparison and are evaluated under identical portfolio construction and cost assumptions.

Table 4: Performance summary for SP3 ($k = 5$), **net of 15 bps per half-turn**. Panels **A–B** correspond to horizons 1D and 7D. Mean is the *daily* average portfolio return (decimal units); Ann.Vol is the **annualized** standard deviation of daily returns (252-day convention); Sharpe/Sortino are **annualized** from daily returns using $\sqrt{252}$; VaR_{5%} and CVaR_{5%} are *daily* tail-risk measures (not annualized). Momentum baselines are recomputed per horizon under the same tranche and cost assumptions. *Bold indicates the maximum Sharpe within each horizon (not statistically significant).*

Panel A: 1D						
Model	Mean	Ann.Vol	Sharpe	Sortino	VaR _{5%}	CVaR _{5%}
LSTM	0.00191	1.67464	0.28727	0.37306	-0.20361	-0.26729
GRU	-0.00585	1.66685	-0.88390	-1.04996	-0.24716	-0.27470
Transformer	0.00189	1.00545	0.47376	0.73534	-0.07517	-0.13617
ResNet	-0.00778	0.74614	-2.62750	-2.82634	-0.06601	-0.14711
DenseNet	-0.00641	0.71080	-2.27175	-2.54989	-0.05254	-0.12491
ViT	-0.00470	0.56778	-2.08787	-2.51987	-0.04316	-0.09984
Momentum(7)	-0.01306	1.50399	-2.18786	-2.63262	-0.21423	-0.27014
Momentum(14)	-0.01281	1.49433	-2.15985	-2.70818	-0.19798	-0.25503
Momentum(28)	-0.02180	1.55700	-3.52839	-4.18047	-0.24070	-0.26883
Panel B: 7D						
Model	Mean	Ann.Vol	Sharpe	Sortino	VaR _{5%}	CVaR _{5%}
LSTM	0.00665	1.62633	1.02972	1.53822	-0.17207	-0.22653
GRU	-0.00070	1.67271	-0.10499	-0.12680	-0.23202	-0.26148
Transformer	0.00675	1.33948	1.26904	1.79527	-0.14416	-0.19405
ResNet	-0.00093	0.54427	-0.43171	-0.52093	-0.04174	-0.07646
DenseNet	-0.00060	0.82348	-0.18487	-0.20909	-0.05572	-0.13537
ViT	-0.00025	0.52009	-0.11911	-0.18413	-0.04348	-0.06986
Momentum(7)	-0.00448	1.44790	-0.77926	-0.96065	-0.19947	-0.24512
Momentum(14)	-0.00755	1.47362	-1.29099	-1.60684	-0.19110	-0.24524
Momentum(28)	-0.00831	1.63588	-1.28082	-1.55908	-0.20739	-0.27118

Table 4: (*continued*) Performance summary for SP3 ($k = 5$), **net of 15 bps per half-turn**. **Panels C–D** correspond to horizons 14D and 28D. Definitions as in Panels A–B.

Panel C: 14D						
Model	Mean	Ann.Vol	Sharpe	Sortino	VaR _{5%}	CVaR _{5%}
LSTM	-0.00107	1.66322	-0.16238	-0.21966	-0.21357	-0.24496
GRU	0.00165	1.54076	0.27007	0.33006	-0.21544	-0.24557
Transformer	0.00346	1.27447	0.68461	0.94065	-0.15137	-0.19966
ResNet	0.00332	0.66745	1.25422	1.71148	-0.04502	-0.09075
DenseNet	0.00032	0.83169	0.09738	0.10620	-0.05352	-0.13847
ViT	0.00094	0.50379	0.47196	0.65115	-0.03346	-0.06668
Momentum(7)	-0.00059	1.37659	-0.10809	-0.13716	-0.17790	-0.22400
Momentum(14)	-0.00185	1.44097	-0.32370	-0.39854	-0.17539	-0.24023
Momentum(28)	-0.00886	1.42166	-1.57090	-1.82401	-0.17988	-0.25639
Panel D: 28D						
Model	Mean	Ann.Vol	Sharpe	Sortino	VaR _{5%}	CVaR _{5%}
LSTM	0.00359	1.64682	0.54927	0.70835	-0.20305	-0.23733
GRU	-0.00543	1.60757	-0.85043	-0.92327	-0.24458	-0.27238
Transformer	0.00735	1.36727	1.35552	1.84633	-0.16984	-0.20379
ResNet	0.00164	0.54486	0.75793	1.01577	-0.04650	-0.07436
DenseNet	0.00069	0.79408	0.22019	0.22226	-0.04557	-0.12941
ViT	0.00322	0.52304	1.55139	2.84508	-0.04133	-0.05866
Momentum(7)	-0.00027	1.49090	-0.04537	-0.05639	-0.19343	-0.23090
Momentum(14)	-0.00122	1.33892	-0.22939	-0.29138	-0.16762	-0.21933
Momentum(28)	-0.00496	1.43401	-0.87221	-1.00560	-0.18374	-0.24557

Table 4 yields three implications regarding the economic relevance of deep learning-based portfolio signals in SP3. First, net-of-cost performance at short horizons is weak even when average returns are occasionally positive. At the 1-day horizon, LSTM and Transformer deliver positive mean daily returns of 0.00191 and 0.00189, respectively (about 0.19%), yet their annualized Sharpe ratios remain below 0.50 after costs, while ResNet, DenseNet, and ViT exhibit sharply negative Sharpe ratios of -2.63, -2.27, and -2.09. Second, performance improves modestly as the forecast horizon lengthens, consistent with lower mechanically induced turnover under horizon-matched tranches. At the 14-day horizon, ResNet achieves an annualized Sharpe ratio of 1.25 and Transformer reaches 0.68, and at the 28-day horizon Transformer and ViT reach 1.36 and 1.55, respectively. However, these positive cases are not uniform across models and horizons (for example, Transformer drops to 0.47 at 1D and 0.68 at 14D). Third, cross-model differences in volatility are economically meaningful but do not translate into robust risk-adjusted gains. Image-based models exhibit substantially lower annualized volatility (often below 0.60, for example, ViT 0.52–0.57 and ResNet 0.54–0.75) than several sequence-based models (often above 1.20, for example, LSTM 1.63–1.67 and GRU 1.54–1.67), yet downside risk remains sizable: daily 5% VaR is frequently worse than -4% for multiple horizons (e.g., LSTM -17% to -21% and Transformer -14% to -17% at longer horizons). Overall, Table 4 indicates that weak portfolio-level predictability does not reliably translate into economically meaningful net performance once realistic transaction costs are applied.

Figure 2 visualizes the cumulative NAV paths in SP3 corresponding to the net performance in Table 4 and highlights how gains and losses evolve over time under horizon-matched implementation. At the 1-day horizon, NAV trajectories are highly unstable, with early gains quickly reversed and prolonged drawdowns thereafter, consistent with the difficulty of sustaining performance under frequent trading. At intermediate horizons, some models exhibit temporary recoveries, but these are not persistent and tend to deteriorate during adverse regimes, including the 2022 downturn. At longer horizons (14 and 28 days), NAV paths become smoother and less erratic especially for image-based models, yet improvements remain episodic rather than steadily compounding. Overall, Figure 2 reinforces that positive net performance, when it appears, is concentrated in limited subperiods and remains sensitive to regime shifts.

In sum, the SP3 results suggest that any net performance advantages are fragile and may be driven by sample specific dynamics rather than persistent structure. To distinguish genuine economic content from sampling variability and data snooping effects, the next subsection subjects the findings to formal significance tests and robustness checks, including paired block bootstraps with horizon-matched blocks, multiple testing adjustments, and evaluations across alternative study periods and portfolio breadths.

3.3. Robustness Tests

We examine the robustness and statistical significance of the portfolio-level results documented in Section 2 and 3. Specifically, we assess robustness along

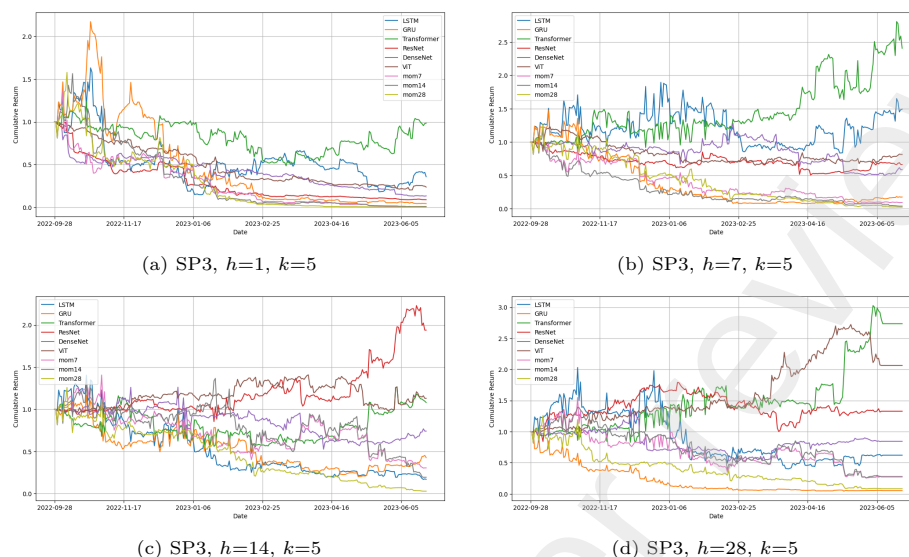


Figure 2: Cumulative NAV for SP3 across horizons ($k=5$), equal-dollar long-short with horizon-matched tranches, **net of 15 bps per half-turn**. Each series is rebased to 1 at the start of the test window.

three dimensions. First, we test the statistical significance of portfolio returns and risk-adjusted performance using paired block bootstrap procedures with block lengths matched to the forecast horizon, thereby accounting for serial dependence induced by overlapping holding periods. Second, we conduct pairwise comparisons across deep learning architectures and benchmark strategies, applying multiple-testing adjustments to control for data-snooping bias. Third, we examine the sensitivity of the results to alternative study periods and portfolio breadths, verifying that the qualitative conclusions are not driven by a particular time window or position size. Collectively, these tests provide a disciplined evaluation of whether portfolio-level predictability persists beyond the representative SP3 case and whether any apparent economic gains can be considered robust.

Table 5 reports pairwise tests of differences in risk-adjusted performance across deep learning architectures for SP3, focusing on the 28-day forecast horizon where net Sharpe ratios are relatively higher. Differences in Sharpe ratios are evaluated using paired block bootstrap procedures with block lengths matched to the holding horizon, and p-values are adjusted for multiple testing using the Benjamini–Hochberg false discovery rate (FDR) procedure. Across all pairwise comparisons, none of the adjusted p-values falls below conventional significance thresholds. In particular, models that exhibit comparatively higher Sharpe ratios in Table 4 such as sequence-based Transformer or the Vision Transformer do not statistically dominate alternative architectures once sampling variability and multiple-testing effects are taken into account. A pooled

28D matrix across SPs appears in Appendix C4, and all other SP–horizon combinations (and momentum-including matrices) are omitted for brevity.

Table 5: Pairwise tests of Sharpe-ratio differences (ΔSharpe) among deep models in SP3 ($k=5$, 28D horizon). The upper triangle reports adjusted two-sided p -values for $\Delta\text{Sharpe} = \text{Sharpe(A)} - \text{Sharpe(B)}$; the diagonal entries are “—”. All p -values are obtained via paired block bootstraps (28-day blocks) on daily simple returns and adjusted using the Benjamini–Hochberg FDR procedure (10%). *All adjusted p -values exceed 0.10; no statistically significant pairwise differences.*

	LSTM	GRU	Transformer	ResNet	DenseNet	ViT
LSTM	—	0.835	0.905	0.863	0.887	0.906
GRU		—	0.906	0.998	0.972	0.953
Transformer			—	0.967	0.954	0.963
ResNet				—	0.978	0.989
DenseNet					—	0.965
ViT						—

We next examine whether the portfolio-level patterns documented for SP3 generalize across alternative market environments. To this end, we replicate the analysis for the remaining study periods (SP1–SP5), which collectively span distinct market regimes, including prolonged bull markets, sharp drawdowns, and periods of heightened volatility. Across study periods, the qualitative findings remain stable. Both portfolio hit rates and rank information (IC) modestly improve as the forecasting horizon lengthens. Specifically, the sequence-based Transformer is the most consistent on rank information at 14–28 days, while image-based models (ViT/ResNet) tend to deliver steadier NAV paths via lower annualized volatility. However, net economic performance remains fragile once transaction costs are incorporated. Importantly, no deep learning architecture consistently delivers positive and statistically significant net performance across all study periods. Instances of relatively favorable Sharpe ratios are episodic and vary by horizon and period, reinforcing the conclusion that apparent economic gains are not regime-invariant but instead depend on specific market conditions.

We further assess robustness with respect to portfolio breadth by varying the number of assets for $k \in \{1, 2, 5, 10, 20\}$. Expanding the portfolio breadth generally reduces return volatility and tail risk, as expected from increased diversification. However, this reduction in risk does not materially alter the core economic conclusions. While larger portfolios produce smoother return profiles, average returns and Sharpe ratios do not improve in a systematic or economically meaningful way. Conversely, narrower portfolios amplify return dispersion and volatility but fail to generate robust improvements in risk-adjusted performance. Across all breadth configurations, the relative ordering of model families remains largely unchanged, and no specification consistently overcomes the erosion of returns induced by turnover and transaction costs.⁶

⁶We report (i) cross-SP medians and interquartile ranges for hit rate, IC, and Sharpe by

4. Discussion

This study set out to compare two representation families, image-based (ResNet, DenseNet, ViT) and sequence-based (LSTM, GRU, Transformer), for cross-sectional cryptocurrency prediction, and to assess whether their signals translate into economically meaningful long-short portfolios under realistic trading frictions. The results reveal a consistent pattern: while deep learning models extract weak but systematic cross-sectional information at the portfolio level, this predictability rarely translates into robust economic value once turnover and transaction costs are taken into account. Lower effective trading costs or coarser execution clocks may partially mitigate the tradability gap documented here; however, under daily implementation and realistic fee assumptions, transaction costs remain a binding constraint on economic performance.

A central insight of the results is the distinction between pointwise predictability and ordinal learning, and how this distinction depends on data representation. Our classification-based formulation is instrumental in revealing this separation, as it emphasizes relative ordering and signal stability rather than fragile point estimates of returns under heavy-tailed noise. Portfolio directional hit rates remain close to the 0.5 benchmark across models and horizons, indicating limited predictive accuracy at the asset level. At the same time, positive information coefficients, particularly at longer horizons, show that models learn relative ordering information that becomes visible only when signals are aggregated across assets. This divergence is especially pronounced in volatile markets such as cryptocurrencies, where idiosyncratic noise dominates short-horizon outcomes but cross-sectional structure can persist over longer holding periods.

These patterns reflect distinct learning behaviors induced by the choice of representation. Sequence-based representations preserve temporal ordering and magnitude information in OHLCV data, enabling models—particularly Transformers—to aggregate dependencies over longer horizons and to strengthen cross-sectional ranking as the forecast horizon increases. Image-based representations, by contrast, compress temporal dynamics into spatial patterns and impose stronger inductive biases through visual abstraction. This transformation regularizes short-term noise and leads to smoother portfolio trajectories, albeit without systematically improving ranking quality. Viewed through this lens, the portfolio results are best interpreted not as a contest between architectures, but as evidence that representation choice primarily shapes the character of learned signals—persistence versus smoothness—before economic constraints determine their ultimate tradability.

Despite this portfolio-level predictability, the economic value of learned signals largely disappears once realistic trading frictions are imposed. The horizon-matched tranche implementation reveals that turnover is structurally tied to

model-horizon, (ii) k -breadth sensitivities for key metrics, and (iii) turnover summaries by model and horizon, which corroborate the conclusions above.

the forecast horizon, with baseline gross turnover equal to $2/h$ per day. At short horizons, this turnover is sufficiently large that even modest transaction costs overwhelm expected gains, yielding negative net performance despite occasional positive average returns. As horizons lengthen and mechanically induced turnover declines, net performance improves modestly, and a small number of model-horizon combinations exhibit Sharpe ratios exceeding unity. However, these gains are episodic, horizon-specific, and sensitive to market regimes, as illustrated by the cumulative NAV paths in Figure 2. Performance improvements are concentrated in limited subperiods rather than accruing steadily over time, and drawdowns remain economically meaningful even in cases with favorable risk-adjusted metrics.

These findings highlight that the predictability-tradability gap in cryptocurrency markets is not simply a consequence of overly conservative cost assumptions, but reflects a structural mismatch between signal strength and implementation frictions. The learned signals are real in a statistical sense, but their magnitude is small relative to the costs required to extract them through frequent rebalancing. Moreover, the absence of statistically significant dominance in pairwise model comparisons suggests that no particular representation or architecture consistently resolves this mismatch. What differentiates models in practice is less their ability to generate economically robust alpha than the way their inductive biases shape volatility, turnover sensitivity, and exposure to regime shifts.

Our study carries several methodological implications for financial data science. First, they underscore the importance of separating predictive learning from decision-making. Improvements in classification accuracy or loss-based metrics are insufficient indicators of economic value unless evaluated within a decision framework that reflects execution constraints. Second, the results highlight the need for cost-aware learning. Standard training objectives are agnostic to transaction costs and turnover, and may therefore optimize statistical criteria that are misaligned with economic performance. Incorporating trading frictions, turnover penalties, or signal persistence directly into learning objectives represents a promising avenue for aligning prediction with implementation. Third, the comparison of representations suggests that representation learning primarily affects stability and risk characteristics, rather than the magnitude of achievable alpha. More expressive architectures do not, by themselves, resolve the fundamental mismatch between weak signals and trading frictions.

More broadly, the results caution against interpreting incremental improvements in predictive benchmarks as evidence of economic viability. In financial settings with substantial frictions, the marginal value of predictive accuracy is highly context dependent and must be assessed jointly with execution constraints, portfolio construction rules, and risk management considerations. This implies a shift in emphasis from better prediction to better integration of prediction with decision-making. Models should be evaluated not only on their ability to learn patterns in data, but also on how those patterns survive contact with the economic environment in which decisions are implemented.

5. Conclusion

This paper investigates the relationship between statistical predictability and economic tradability under realistic frictions in cryptocurrency markets by comparing image-based and sequence-based deep learning models. Using five rolling study periods (SP1–SP5), 30-day lookbacks, multiple forecasting horizons (1, 7, 14, 28 days), and a consistent tranche-based backtesting framework with 15 bps per half-turn costs, we provided a unified assessment of predictability and tradability across architectures.

The results yield a clear conclusion. Deep learning models can extract weak but systematic cross-sectional information, as evidenced by portfolio-level directional signals and positive rank information coefficients. However, this statistical predictability does not reliably convert into robust economic value. Under horizon-matched portfolio construction with explicit transaction costs, net performance is fragile, episodic, and highly sensitive to turnover and market regimes. No model architecture or data representation consistently dominates once proper statistical inference and robustness checks are applied.

Overall, the results underscore a fundamental distinction between learning predictive patterns and exploiting them in practice. In markets characterized by high volatility, regime instability, and nontrivial execution costs, improvements in predictive metrics alone are insufficient indicators of tradable opportunity. The predictability–tradability gap observed in this study reflects a structural mismatch between the magnitude of learned signals and the frictions required to extract them through frequent rebalancing, rather than a failure of deep learning methods to capture statistical structure.

While the analysis focuses on a specific set of models, cryptocurrency markets, and implementation choices, the conclusions point toward several directions for future research. Incorporating cost-aware or decision-aware learning objectives, exploring alternative execution frequencies and portfolio constructions, and integrating richer data sources such as order-book or on-chain information may help improve the alignment between prediction and implementation. At the same time, any such extensions must be evaluated within disciplined frameworks that account for turnover, transaction costs, and statistical uncertainty.

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Appendix A. Excluded Coins and Data Filters

This appendix lists excluded assets and applied data filters. The investable universe is defined as the top 100 assets by market capitalization on the first day of each training window, excluding fiat-pegged stablecoins and assets with missing or non-finite inputs.

Table A1: Overview of coins excluded from the study and reasons for exclusion.

Coin Name	Coin Symbol	Exclusion Reason
Vechain [Token]	VEN	Data issues
ThoreNext	THX	Data issues
ZB Token	ZB	Data issues
ThoreCoin	THR	Data issues
EDUCare	ETK	Data issues
Dimension Chain	EON	Data issues
Karatgold Coin	KBC	Data issues
Silverway	SLV	Data issues
GAPS	GAP	Data issues
Bitbook Gambling	BXK	Data issues
Dynamic Trading Rights	DTR	Data issues
Flexacoin	FXC	Data issues
Newton	NEW	Data issues
VestChain	VEST	Data issues
Aave	LEND	Data issues
Counos X	CCXX	Data issues
BitcoinPoS	BPS	Data issues
Hyperion	HYN	Data issues
Zelwin	ZLW	Data issues
The Midas Touch Gold	TMTG	Data issues
Elrond	ERD	Data issues
Largo Coin	LRG	Data issues
BeautyPayCoin	BPC	Data issues
Idea Chain Coin	ICH	Data issues
botXcoin	BOTX	Data issues
Velas	VLX	Data issues
Klaytn	KLAY	Data issues
TerraUSD	UST	Data issues
Revain	REV	Data issues
MEXC Token	MEXC	Data issues
ICO COIN	INO	Data issues
Gifto	GTO	Data issues
Bitcoin Private	BTCP	Data issues
Stratis	STRAT	Data issues
RChain	RHOC	Data issues

(Continued on next page)

Table A1 (continued)

Coin Name	Coin Symbol	Exclusion Reason
Golem	GNT	Data issues
Mixin	XIN	Data issues
Substratum	SUB	Data issues
Cryptonex	CNX	Data issues
Storm	STORM	Data issues
GXChain	GXC	Data issues
SALT	SALT	Data issues
Veritaseum	VERI	Data issues
Zcoin	XZC	Data issues
Nucleus Vision	NCASH	Data issues
BTMX	BTMX	Data issues
Paxos Standard	PAX	Data issues
Linkey	LKY	Data issues
Project Pai	PAI	Data issues
MaidSafeCoin	MAID	Data issues
ODEM	ODE	Data issues
Insight Chain	INB	Data issues
Single Collateral DAI	SAI	Data issues
Revain	R	Data issues
Pundi X	NPXS	Data issues
PayPie	PPP	Data issues
Tether	USDT	Stablecoin
STASIS EURO	EURS	Stablecoin
Dai	DAI	Stablecoin
Binance USD	BUSD	Stablecoin
HUSD	HUSD	Stablecoin
USD Coin	USDC	Stablecoin
TrueUSD	TUSD	Stablecoin
Gemini Dollar	GUSD	Stablecoin

Appendix B. Supplemental Figures

Appendix B presents cumulative NAV trajectories for each study period (SP1–SP5) and all model families (LSTM, GRU, Transformer, ResNet, DenseNet, ViT) across horizons $h \in \{1, 7, 14, 28\}$. Study Period 3 corresponds to Figure 2 in the main text, and all NAVs are generated under identical portfolio and cost conventions: equal-dollar long–short portfolios with horizon-matched tranches, net of 15 bps per half-turn, rebased to 1 at the start of each test window.

Appendix C. Supplemental Tables

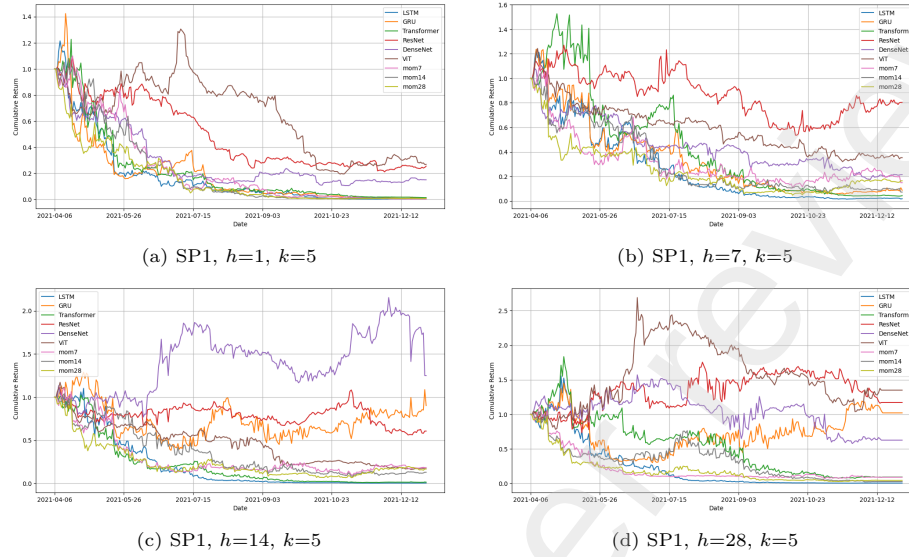


Figure B1: Cumulative NAVs for SP 1 across horizons ($k=5$). Equal-dollar long-short portfolios with horizon-matched tranches, net of 15 bps per half-turn. Each series is rebased to 1 at the start of the test window; y -axis limits are kept consistent across panels.

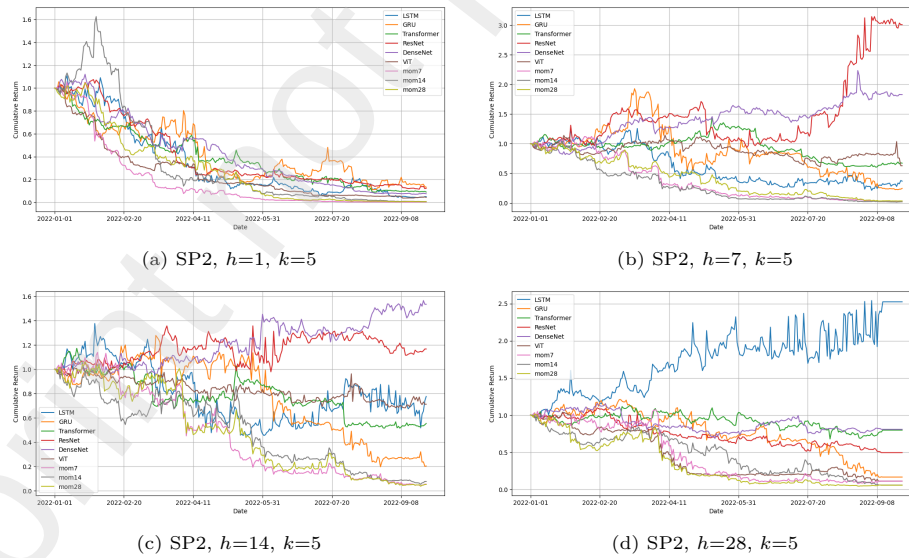


Figure B2: Cumulative NAVs for SP 2 across horizons ($k=5$). Portfolio and visual settings are identical to Figure B1.

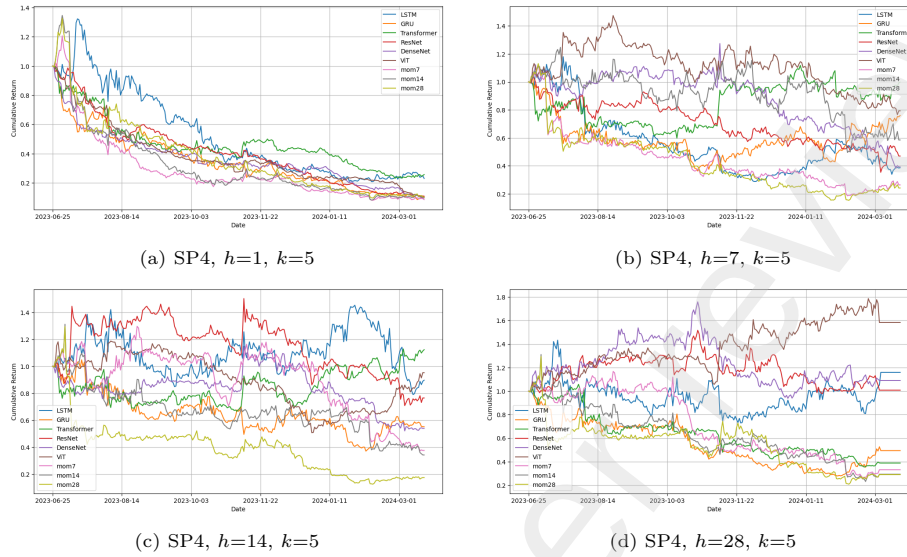


Figure B3: Cumulative NAVs for SP 4 across horizons ($k=5$). Portfolio and visual settings are identical to Figure B1.

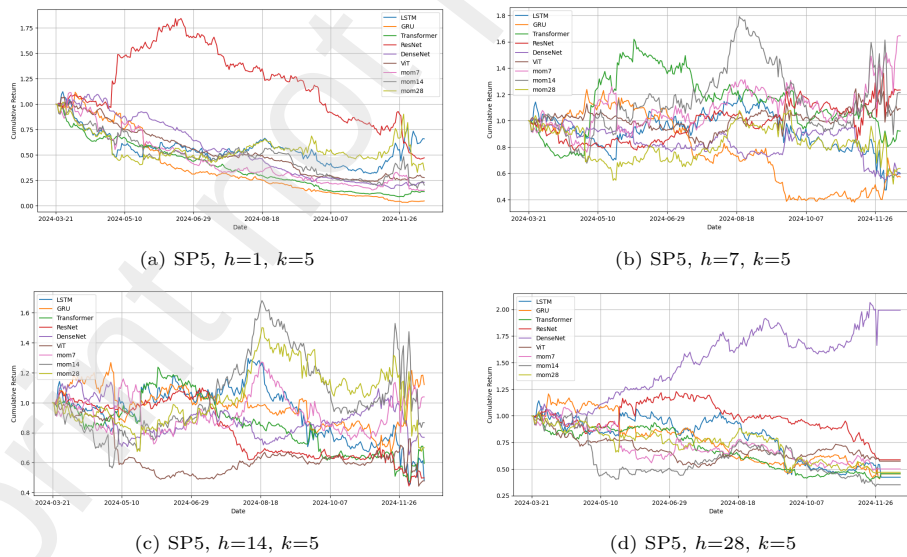


Figure B4: Cumulative NAVs for SP 5 across horizons ($k=5$). Portfolio and visual settings are identical to Figure B1.

Table C1: Cross-SP medians (and IQR) of portfolio directional hit-rate by model and horizon at $k=5$. Hit rates are computed on traded names only (top- k longs and bottom- k shorts), treating neutrals as incorrect, under horizon-matched tranches and 15 bps per half-turn costs. Values are summarized across the five study periods (SP1–SP5).

	1D	7D	14D	28D
LSTM	0.434 [0.013]	0.447 [0.006]	0.444 [0.018]	0.431 [0.024]
GRU	0.514 [0.060]	0.498 [0.028]	0.470 [0.015]	0.456 [0.019]
Transformer	0.437 [0.036]	0.482 [0.017]	0.469 [0.026]	0.465 [0.037]
ResNet	0.454 [0.041]	0.482 [0.009]	0.472 [0.006]	0.443 [0.016]
DenseNet	0.441 [0.027]	0.474 [0.025]	0.477 [0.007]	0.443 [0.022]
ViT	0.454 [0.029]	0.481 [0.004]	0.466 [0.009]	0.438 [0.010]

Table C2: Cross-SP medians (and IQR) of Spearman IC by model and horizon at $k=5$. ICs are computed daily as cross-sectional Spearman correlations between the score $s = \hat{p}_{\text{up}} - \hat{p}_{\text{down}}$ and h -ahead simple returns, then averaged over the five study periods (SP1–SP5).

	1D	7D	14D	28D
LSTM	-0.0492 [0.0733]	0.0059 [0.0462]	-0.0017 [0.0570]	-0.0061 [0.0678]
GRU	0.0920 [0.0328]	0.0545 [0.0358]	0.0424 [0.0416]	0.0403 [0.0543]
Transformer	0.0193 [0.0163]	0.0460 [0.0089]	0.0547 [0.0500]	0.0755 [0.0738]
ResNet	0.0233 [0.0100]	0.0155 [0.0124]	0.0167 [0.0108]	0.0020 [0.0266]
DenseNet	0.0263 [0.0120]	0.0383 [0.0277]	0.0387 [0.0227]	0.0045 [0.0364]
ViT	0.0258 [0.0033]	0.0201 [0.0061]	0.0154 [0.0132]	-0.0101 [0.0236]

Table C3: Cross-SP medians (and IQR) of daily net Sharpe by model and horizon at $k=5$. Sharpe ratios are computed on daily simple returns under horizon-matched tranches and 15 bps per half-turn transaction costs, then summarized across the five study periods (SP1–SP5).

	1D	7D	14D	28D
LSTM	-0.0597 [0.0758]	-0.0123 [0.0531]	-0.0102 [0.0415]	0.0346 [0.0890]
GRU	-0.1143 [0.0878]	-0.0066 [0.0182]	0.0170 [0.0454]	-0.0464 [0.0189]
Transformer	-0.1007 [0.0113]	0.0087 [0.0549]	-0.0252 [0.0734]	-0.0248 [0.0646]
ResNet	-0.1534 [0.0732]	0.0112 [0.0672]	-0.0080 [0.0483]	0.0145 [0.0892]
DenseNet	-0.1832 [0.0498]	-0.0484 [0.0615]	0.0061 [0.0571]	0.0177 [0.0119]
ViT	-0.1834 [0.0998]	-0.0120 [0.0122]	-0.0179 [0.0538]	0.0474 [0.1481]

Table C4: Adjusted p -values summary at $k=5$ (BH-FDR at 10%). Two-sided tests of mean daily returns (paired block bootstraps; block length h). Panel (a) shows tests *against zero*; tests against momentum baselines (7/14/28-day) are also non-significant and omitted for brevity. Panel (b) reports pairwise Δ Sharpe p -values at $h=28$ (upper triangle) pooled across SPs.

Panel (a): Checkmarks indicate $q \leq 0.10$ after BH-FDR (vs zero)						
	1D	7D	14D	28D		
LSTM						
GRU						
Transformer						
ResNet						
DenseNet						
ViT						

Note: All BH-FDR adjusted q -values exceed 0.10 in Panel (a).

Panel (b): Pairwise Δ Sharpe p -values at 28D (upper triangle), BH-FDR adjusted						
	LSTM	GRU	Transformer	ResNet	DenseNet	ViT
LSTM	–	1.000	1.000	1.000	1.000	1.000
GRU		–	1.000	1.000	1.000	1.000
Transformer			–	1.000	1.000	1.000
ResNet				–	1.000	1.000
DenseNet					–	1.000
ViT						–

Table C5: Cross-SP medians (and IQR) of daily net Sharpe by portfolio breadth $k \in \{1, 2, 5, 10, 20\}$ at the 28-day horizon. Sharpe ratios are computed on daily simple returns under horizon-matched tranches and 15 bps per half-turn costs. Results confirm that varying k does not change the qualitative ranking of model families.

Model	$k=1$	$k=2$	$k=5$	$k=10$	$k=20$
LSTM	0.0101 [0.0160]	0.0252 [0.0112]	0.0346 [0.0890]	-0.0009 [0.1395]	0.0251 [0.1630]
GRU	-0.0579 [0.0310]	-0.0278 [0.0164]	-0.0464 [0.0189]	-0.0492 [0.0691]	-0.0158 [0.1147]
Transformer	0.0175 [0.1117]	-0.0040 [0.1158]	-0.0248 [0.0646]	-0.0562 [0.0837]	-0.0242 [0.2224]
ResNet	-0.0626 [0.0616]	-0.0569 [0.1107]	0.0145 [0.0892]	0.0635 [0.0891]	0.0629 [0.2340]
DenseNet	0.0019 [0.0370]	0.0241 [0.0496]	0.0177 [0.0119]	0.0007 [0.0652]	0.0110 [0.1041]
ViT	-0.0117 [0.0609]	-0.0085 [0.0160]	0.0474 [0.1481]	0.0462 [0.1637]	-0.0616 [0.1679]